design, symbolizing the shift from early piston-powered drones to modern, reusable jet-powered UAVs capable of realistic flight simulation and intelligence missions. The Ryan Firebee thus represents a key technological milestone in the evolution of second-generation drones, combining performance, reusability, and mission adaptability.




**Figure 3.4** Ryan Firebee (1951) – the first jet-powered target and reconnaissance drone.⏎

### 3.5.2.2 Ryan Model 147 “Lightning Bug” (1964)

The Ryan Model 147 “Lightning Bug” (1964), developed by Ryan Aeronautical Company (United States), represents one of the most significant advancements in the history of unmanned aerial reconnaissance. Evolving from the successful Ryan Firebee platform, the Lightning Bug was a modified, jet-powered UAV specifically designed for strategic intelligence gathering during the Vietnam War and the Cold War. It was equipped with high-resolution cameras, sensor payloads, and autonomous